

Practical 7

Aim: Implement a stack using arrays in C++, and perform push, pop, and display operations.

Source-Code:

```
#include<iostream>

using namespace std;

#define MAX 5

int stack[MAX];

int top = -1;

void push(int value) {
    if (top == MAX - 1) {
        cout << "Stack Overflow! Cannot push element.\n";
    } else {
        top++;
        stack[top] = value;
        cout << value << " pushed into stack.\n";
    }
}

void pop() {
    if (top == -1) {
        cout << "Stack Underflow! Cannot pop element.\n";
    } else {
        cout << stack[top] << " popped from stack.\n";
        top--;
    }
}

void display() {
    if (top == -1) {
        cout << "Stack is empty.\n";
    } else {
        cout << "Stack elements are: ";
        for (int i = top; i >= 0; i--) {
            cout << stack[i] << " ";
        }
        cout << endl;
    }
}
```

```
    }
}

int choice, value;

int main() {
    do {
        cout << "\n--- Stack Menu ---\n";
        cout << "1. Push\n";
        cout << "2. Pop\n";
        cout << "3. Display\n";
        cout << "4. Exit\n";
        cout << "Enter your choice: ";
        cin >> choice;
        switch (choice) {
            case 1:
                cout << "Enter value to push: ";
                cin >> value;
                push(value);
                break;
            case 2:
                pop();
                break;
            case 3:
                display();
                break;
            case 4:
                cout << "Exiting program.\n";
                break;
            default:
                cout << "Invalid choice! Please try again.\n";
        }
    } while (choice != 4);
    return 0;
}
```

Output:

```
--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter value to push: 10
10 pushed into stack.

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter value to push: 20
20 pushed into stack.

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 1
Enter value to push: 30
30 pushed into stack.

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 3
Stack elements are: 30 20 10

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 2
30 popped from stack.

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 3
Stack elements are: 20 10

--- Stack Menu ---
1. Push
2. Pop
3. Display
4. Exit
Enter your choice: 4
Exiting program.
```